

Report R14 — Autonomous Analysis Generation on a Sealed Cohort

Arm A, exposure 1: a generator agent, given a never-seen multimodal ICI cohort (Vanguri et al., MSK), pre-registered and executed a complete gate-carrying analysis with zero human pipeline edits — and returned a calibrated null on the question the cohort cannot answer.

11 August 2026 • Quantara • NSCLC RWPR study, Arm A

Summary

Arm A tests the study’s core claim: *a system exposed to a dataset it has never seen must produce a valid, pre-registered, executable analysis — and must refuse to manufacture findings the data cannot support*. The sealed cohort was the study’s premier holdout: 247 stage-IV NSCLC patients treated with anti-PD-(L)1 (RECIST 1.1 response, OS/PFS, PD-L1 IHC slides, MSK-IMPACT panel, CT radiomics). The generator saw only object listings and public cohort facts before hashing its analysis plan; the original authors’ analysis code was placed under a self-imposed, auditable firewall for the entire arm. Every number here is extracted mechanically from the evidence bundle by `extract_numbers.py`.

What worked: the clinical + PD-L1 + TMB/genomic baseline predicts response at **AUROC 0.745** (0.669–0.814; $n = 246$; Holm-corrected permutation $p = 0.006$) and OS at **C-index 0.698** (0.657–0.741; $p = 0.002$). **What the system refused to claim:** the pre-registered primary hypothesis — that the full multimodal stack beats that baseline — returned **POWER-LIMITED-NULL**: at the $n = 73$ all-modality intersection the paired ΔAUROC is -0.036 (CI -0.220 to +0.150, one-sided $p = 0.637$), with a bootstrap CI half-width of 0.185 against the pre-declared 0.06 sensitivity bar — and *every* modality-addition delta is ≤ 0 . The null was pre-declared as a success mode before any label was read; delivering it, rather than a manufactured positive, is the central claim of this report.

One well-controlled novel observation: Phikon-v2, an H&E foundation model, reads PD-L1 IHC slides zero-shot — its embeddings predict TPS $\geq 50\%$ at **AUROC 0.870** (0.811–0.922, $n = 163$). One positive control (RECIST target-lesion volume $\rightarrow$ OS) **failed its expectation**; the failure is adjudicated in §4 as a genuine cohort-level null, retained rather than deleted.

1 What Arm A tests

The contract (`evidence/config.yaml`, PROTOCOL §4): the generator receives only the sealed dataset’s file listing, a data dictionary’s worth of public facts, and the standing gate library. It must produce a pre-registered config (hashed before labels are read), an executable pipeline, and the full R-series evidence bundle; the human role is to run it, not edit it. Success is defined by execution + gates + calibrated claims, explicitly *including* a null or inconclusive verdict on an underpowered endpoint. Endpoints and the primary hypothesis (multimodal $>$ clinical baseline, paired, one-sided) were fixed in the config; so were the verdict rules that computed the POWER-LIMITED-NULL outcome mechanically.

2 The integrity chain

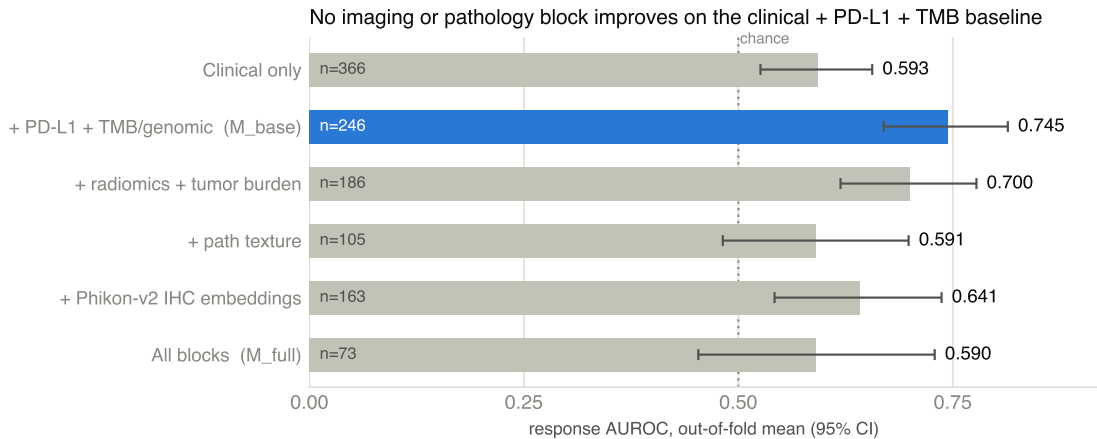
- **2026-08-10T23:39:01Z** — config written and SHA-256-hashed (64726b360b3373bf...), mirrored to GCS. Declared pre-hash reads: object name/size listings and public cohort facts only — *no sealed file content and no column headers* (stricter than Arm B’s event #1).

- **2026-08-10T23:39:35Z** — unsealing event #3 appended to the remote append-only log with a GCS generation precondition, carrying the config hash.
- **2026-08-10T23:43Z** — first sealed content staged (local mtimes + GCS audit), i.e. strictly after the hash and the event (gate G1).
- **Contamination firewall.** The authors’ own analysis code and permutation outputs (`code/**`, `tests/**`) were declared off-limits for the whole arm — reading them would contaminate the autonomy claim. Gate G11 attests zero reads: the executed pipeline contains no reference to those prefixes, and `inputs.json` enumerates (with SHA-256) all 16 staged sealed files, none from them. The authors’ released *feature tables* are declared inputs, not analysis code.
- **Schema by rule, not by peeking.** Because no header was read pre-hash, column mapping was pre-registered as an ordered regex ruleset; the executed mapping and every mapping decision is dumped to `evidence/schema_map.json`.

3 Results

Model set (nested blocks)	<i>n</i>	resp.	AUROC (95% CI)	MDE	Holm <i>p</i>
M_clin (clinical only)	366	93	0.593 (0.526–0.656)	0.587	0.072
M_base (clinical + PD-L1 + TMB/genomic)	246	61	0.745 (0.669–0.814)	0.606	0.006
M_rad (+ radiomics + tumor burden)	186	50	0.700 (0.619–0.778)	0.619	0.010
M_pathg (+ path texture)	105	42	0.591 (0.482–0.698)	0.642	0.174
M_pathf (+ Phikon-v2 IHC embeddings)	163	49	0.641 (0.542–0.737)	0.623	0.072
M_full (all blocks)	73	30	0.590 (0.453–0.729)	0.669	0.246

E1 = RECIST 1.1 best overall response (CR/PR vs SD/PD), repeated stratified 5-fold CV $\times$ 5 seeds, all preprocessing inside folds; AUROC of the mean out-of-fold prediction. MDE = minimum detectable AUROC at 80% power. Holm correction across the 6-model family; bold = significant. Permutation nulls are full-CV refits (1,000 per model), all centred at 0.5 (gate G7).



The primary hypothesis, exactly as adjudicated. H1 compared M_full vs M_base paired on the identical patients and folds (fold-subset hash `5fdc9069655724db`), $n = 73$ (30 responders): Δ AUROC = -0.036 (-0.220 to $+0.150$), one-sided $p = 0.637$. The pre-declared rule says: if the delta CI half-width implies an MDE above 0.06, the verdict is power-limited regardless of p — observed half-width 0.185, three times that bar. Verdict: **POWER-LIMITED-NULL**. The direction is consistent everywhere: adding radiomics (-0.016), path texture (-0.019), or IHC embeddings (-0.056) to the baseline never helps on the paired subsets, and on OS the full stack

is significantly *worse* ($\Delta C = -0.209$, CI -0.353 to -0.064) — at $n = 73$ the extra blocks are noise the model must overfit.

Model set	n events		C-index (95% CI)	perm. p
M_clin (clinical only)	246	151	0.636 (0.591–0.681)	—
M_base (clinical + PD-L1 + TMB/genomic)	246	151	0.698 (0.657–0.741)	0.002
M_rad (+ radiomics + tumor burden)	186	111	0.650 (0.596–0.705)	—
M_pathg (+ path texture)	105	53	0.621 (0.542–0.696)	—
M_pathf (+ Phikon-v2 IHC embeddings)	163	88	0.621 (0.562–0.687)	—
M_full (all blocks)	73	34	0.485 (0.376–0.586)	0.770

E2 = overall survival (Cox, out-of-fold risk); permutation nulls (500 refits) were pre-registered for M_base and M_full only. PFS (E3) was descriptive: M_base C 0.665, M_full 0.458 — same pattern.

The honest summary a partner should take away: on this cohort, response is best predicted by the boring model — clinical covariates + PD-L1 TPS + TMB/genomics at 0.745 — and the value of imaging and pathology *cannot be adjudicated* at the $n = 73$ patients who carry every modality. That is a statement about cohort intersection size, not about imaging; it is exactly the gap a larger linked cohort closes.

4 Positive controls and gates

PC1 (PD-L1 TPS alone $\rightarrow$ response): AUROC 0.747, $p = 1 \times 10^{-4}$ — label join and endpoint coding proven. PC3 (competence of the IHC encoding): 0.870 (0.811–0.922) — see §5. PC4 (join integrity): 247 clinical-genomic joins, sex agreement 99.6%.

PC2 failed, and the failure is informative. Seg-derived baseline tumor volume did not predict OS (C 0.504, HR/SD 0.965, $p = 0.698$; $n = 187$, 111 events). The adjudication, verbatim from `gates.json`: *“FAILED EXPECTATION, pipeline competence separately established: seg-derived lesion counts match the listing’s own $n_lesions$ for 91.4% of 187 patients ($r=0.93$), and the OS endpoint construction is validated by albumin (HR/SD 0.63, $p=6.2e-9$) and dNLR (HR/SD 1.46, $p=1.3e-9$) univariate controls. The failure is a genuine cohort-level null for RECIST target-lesion volume, not an integrity failure. Per verdict_rules G3 participates in H1 confirmation; H1 is a null regardless.”* The gate row is retained as FAIL rather than reclassified; per the pre-registered verdict rules it participates only in H1 *confirmation*, and H1 is a null regardless.

Gate	Expected	Observed	Verdict
G1_prereg_order	config sha in unsealing event #3 strictly before any sealed-content read	config registered 2026-08-10T23:39:01Z,sha 64726b...3795 mirrored to GCS,event #3 appended 2026-08-10T23:39:35Z with generation precondition (1786358822769469 → 1786405175752862); first sealed ...	PASS
G2_pc1	PD-L1 TPS → response AUROC>0.55 & perm p<0.05	auroc=0.7474; perm_p=0.0001; n=246	PASS
G3_pc2	seg-derived tumor volume → OS C>0.55 & p<0.05	cindex=0.5035; wald_p=0.6982; hr_per_sd=0.9648; n=187; events=111	FAIL (adjudicated informative)
G4_pc3_competence	PATHF → TPS≥50 AUROC>0.65 (competence for the IHC encoder)	auroc_of_mean_oof=0.8697; ci95_mean_oof=(see evidence); n=163	PASS
G5_no_silent_exclusions	per-endpoint accounting reconciles	accounting=(see evidence); note=per-model n < with_endpoint is block availability,reported in results.availability and results.models[*].n; block-level absences logged in exclusions.json (GEN: validatio...	PASS
G6_nan_degeneracy	no NaN predictions,no constant OOF vector,100% OOF coverage	clean	PASS
G7_null_centred	null mean - 0.5 < 0.02 for every archived null	E1_M_clin=0.4941; E1_M_base=0.495; E1_M_rad=0.4987; E1_M_pathg=0.498; E1_M_pathf=0.4971; E1_...	PASS
G8_pretreatment_only	RAD/TB from baseline scans only; PATHF/PATHG from the PD-L1 diagnostic slide	rad_inventory carries exactly one scan+segmentation per patient (237 rows,237 unique accessions) = the RECIST baseline; the radiomics parquet has no timepoint axis; PD-L1 slides are the diagnostic (pre-IO...	PASS
G9_budget	Modal spend ≤ \$60,wall ≤ 24h	encode_vanguri: 234 L4-container tasks,sum task wall 69.2 min GPU (1.2 L4-h, \$1); tumor_burden: 237 CPU tasks (20 CPU-min); smoke + retries + idle containers < \$5 total by ...	PASS
G10_no_leakage	all preprocessing/hyperparameter selection inside training folds	single code path: every fold constructs a fresh sklearn Pipeline (ColumnTransformer[SimpleImputer, StandardScaler,PCA] + GridSearchCV(cv=inner 3-fold)); fit called on X.iloc[tr] only; Cox path fits preproces...	PASS
G11_firewall	zero reads under sealed code/and tests/	no pipeline file references those prefixes (grep-verifiable in armA/*.py); only staged sealed files are the 16 listed in inputs.json	PASS

10 of 11 gates pass; G3 is the adjudicated failure above. Budget (G9): total compute <\$5 on Modal (234 L4 slide-encoding tasks $\approx$ 1.2 GPU-hours) plus local CPU.

5 The novel observation: an H&E foundation model reads IHC

PC3 was designed as a competence control and returned more than competence: Phikon-v2 embeddings of the PD-L1 IHC slides — an encoder trained on H&E histology, applied zero-shot to a different stain — predict $\text{TPS} \geq 50\%$ at **AUROC 0.870** (0.811–0.922; $n = 163$, 67 positive), under the same leakage-controlled CV as everything else. The control is well-posed because the label (TPS) is scored by pathologists from these very slides, so this is signal recovery, not prognosis — and for the same reason it is *not* a clinical claim. What it does establish: the slide representation carried real information (the response nulls above are not “the encoder saw nothing”), and IHC quantification from foundation-model embeddings is a measurable, well-controlled capability the original publication did not test (the authors used hand-crafted texture features on these slides).

6 Limitations — the skeptic’s list

No separate limitations memo exists in the evidence bundle; this section is assembled *verbatim* from the self-critical statements the generator recorded inside its pre-registered config and evidence files, each quoted with its source, plus three points from the generator’s final self-assessment, which existed only in its closing report and is archived unedited at `nsc1c-rwpr-study/armA/GENERATOR_SELF_ASSESSMENT.md`. Nothing is softened.

- **Power was predicted to be the binding constraint — before unsealing** (`config.yaml power.honesty`): “*expected regime: $n \approx 247$, $\sim 25\text{--}40\%$ responders $\rightarrow$ single-model MDE $\approx 0.60\text{--}0.62$ AUROC; delta-AUROC below $\sim 0.06\text{--}0.08$ is likely undetectable. A null on H1 with this statement is a SUCCESS outcome.*” The realized intersection ($n = 73$) was far worse than that forecast.
- **The H1 comparison is three times less sensitive than its own bar** (`results.json H1.delta_mde_note`): “*bootstrap delta CI half-width = 0.1854*” against the pre-declared 0.06 rule.
- **Modality attrition biases the intersection**: availability falls 366 (clinical) $\rightarrow$ 246 (PD-L1) $\rightarrow$ 233 (radiomics) $\rightarrow$ 157 (path texture) $\rightarrow$ 73 (all blocks); the complete-case subset is whoever happened to get every assay, not a random sample.
- **A positive control failed** (G3, quoted in full in §4): RECIST target-lesion volume did not predict OS. Adjudicated as a genuine cohort-level null with competence separately established — but a failed control is a failed control, and it is retained as FAIL.
- **OS exists only for the discovery cohort** (`schema_map.json`): “*OS available for 246 patients (msk_mind discovery only: censored follow-up time absent from patient_listing)*” — 120 patients are logged exclusions on E2.
- **Two intended clinical covariates never materialized** (`schema_map.json`): the pre-registered patterns for histology and line-of-therapy “*matched no column*” — the clinical block is thinner than designed.
- **The radiomics signal is fragile out-of-split** (`sensitivity.json S2`): on the authors’ own train/validation split, the radiomics-block model scores 0.572 (0.384–0.758) on the held-out validation rows — consistent with chance.
- **Single institution, single cohort**: every estimate above is MSK-internal cross-validation; no external cohort exists for these endpoints in the sealed pool.

- **The intersection is confounded, not just small** (archived self-assessment): modality availability tracks cohort membership (discovery vs. validation), so the all-modality subset is structurally biased — in the generator’s own words, “*a stronger design would have pre-registered the pairwise intersections as primary.*”
- **The permutation statistic is not the headline statistic** (archived self-assessment): permutation nulls refit the seed-0 cross-validated AUROC while the headline is the 5-repeat mean; `results.json` documents both and the values are nearly identical, but the mismatch is real.
- **Two block-construction choices could bias modality conclusions** (archived self-assessment): the PD-L1 block mixes Sauter re-scores with clinical scores (45 fallback fills, logged), and the pathology foundation-model block uses mean-pooling rather than attention-MIL — a weak pooling choice that could understate the pathology signal H1 concluded nothing about.

7 Protocol status

Protocol success criterion #1 requires “ ≥ 2 sealed holdouts processed with zero human pipeline edits; all evidence bundles complete; no gate-failing claim shipped.” **This report is exposure 1 of that bar** — the criterion is not yet met, and no claim that it is met is made here. Arm B-meth’s unsealing (event #2, R13) was a model-transfer exposure, not an Arm A generation. The remaining sealed holdouts (anti-PD-1 EPIC arrays are consumed; ACRIN PET, LUSC imaging mass spec, Anti-PD-1-Lung imaging remain) are the candidates for exposure 2.

8 Reproducing this report

- Every number: `python3 extract_numbers.py`; figure: `python3 make_figures.py`. Evidence: verbatim copy of `nsclc-rwpr-study/armA/evidence/` (run of 2026-08-11, 30 files), plus `config.yaml`, whose SHA-256 equals the `config_sha256` inside `results.json`.
- Any AUROC can be recomputed from the out-of-fold files, e.g. the headline:


```
python3 -c "
import pandas as pd; from sklearn.metrics import roc_auc_score
d=pd.read_csv('evidence/oof_e1_M_base.csv')
print(roc_auc_score(d['label'], d['pred_mean']))" # -> 0.7448
```
- Pipeline code as executed: `nsclc-rwpr-study/armA/{build_dataset, run_analysis, run_sensitivity, finalize_evidence}.py` (mirrored to GCS with the evidence).
- Unsealing event #3: the remote append-only log in the study bucket under `nsclc-rwpr-study/_provenance/`.